Optik 114, No. 2 (2003) 69–75
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Optik Optics
International Journal for Light and Electron Optics

Digital audio signal encryption with a virtual optics scheme

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Abstract: A novel approach for enciphering digital audio signal is presented in this article. The operating principle of the approach is on the basis of a virtual optics scheme whereby we apply both virtual wavelength and virtual diffraction distance in conjunction with a complex-valued random mask to design multiple-locks and multiple-keys in the course of audio data encryption and decryption. The audio signal is converted into a sound map, thus the encoded sound map can be encrypted as done for an image. Numerical experiment results show that proposed method is with high sensitivity to parameter discrepancy of virtual optical scheme. The correlation analysis of decrypted audio signal in comparison with original signal is also performed to verify the capability of presented virtual optics scheme for audio signal encryption.

Key words: Digital audio signal – encryption – virtual parameter – digital optics

1. Introduction

Data encryption plays a very important role in the field of information hiding and information security. The study of communications security includes not just encryption but also traffic security, whose essence lies in hiding information. This discipline includes such technologies as: spread spectrum radio, which is widely used in tactical military systems to prevent transmitters being located, temporary mobile subscriber identifiers, used in digital phones to provide users with some measure of location privacy, and many more. Until recently, information hiding techniques received much less attention from the research community and from industry than cryptography, but this is changing rapidly due to the fast development of digital medias and digital right management system [1]. Watermarks (hidden copyright messages) and fingerprints (hidden serial numbers) are all sub-disciplines of information hiding and both fields have been received extensive research recently [2].

Received 4 October 2002; accepted 18 January 2003.

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The general model of hiding data in other data can be described as follows. The embedded data are the message that one wishes to send secretly. It is usually hidden in an innocuous message referred to as a cover text, cover image, or cover audio as appropriate, producing the stegotext or other stego-object. A stego-key is used to control the hiding process so as to restrict detection and/or recovery of the embedded data to parties who know it (or who know some derived key value). As the purpose of steganography is having a covert communication between two parties whose existence is unknown to a possible attacker, a successful attack consists in detecting the existence of this communication [3].

Recently, it has been recognized that the concept and methodology of optical information processing will bring a new generation technology for the data encryption [4]. Various physical-optics based methods for high dimensional data encryption have been proposed in past few years to increase the security level [5]–[7]. More recently, virtual-optics based methods have also been proposed for enhancing the flexibility as opposed to physical-optic based methods and proved to be useful tools for the image encryption [9]–[10], where the plaintext or host image is able to be enciphered using such a methodology with high security level.

In this paper, we present a technique to encipher digital audio signal (DAS) by use of the parameters of a virtual optical scheme. This method is on the basis of wavelength and position sensitivity of Discrete Fresnel Diffraction (DFD) in conjunction with a complex-valued random covering mask. The basic idea behind is motivated from a space-variant Fresnel transform optical processor (SVFTOP) [11]–[12], where the output depends on both the shape and the position of the input function in comparison with reference function. Such a processor is space variant because the Fresnel diffraction introduces spatially dependent phase shifts to the input object. The SVFTOP was originally proposed for optical pattern recognition. Because of the spatially dependent Fresnel shifts, the correlation signal is obtained only when the input object is in the correct location and when it matches the object that is used to generate the reference mask in the filter plane. However, here we only take the advantage of position

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sensitive feature of this kind of structure for the purpose of information hiding. A phase random mask was used in the SVFTOP structure in order 4 to redistribute optical energy on the filter plane [13]–[14]. In contrast, we utilize a complex-valued random mask (RM) as specific cover image in a virtual optical scheme for encrypting the host signal. The rest parts of this paper are organized as follows: Section 2 describes the principle of proposed method; Section 3 presents numerical experiments by taking into account virtual parameters in conjunction with the random mask as tools to design multiple lock and keys for audio signal encryption. The parameter sensitivity of virtual scheme will also be estimated with given structure. Section 4 gives an evaluation for the capability of the scheme through a correlation analysis. The last Section will summarize the major results obtained by this study.

2. Description of the method

The virtual optical scheme used for audio data encryption is illustrated in fig. 1 where the information sheet is an encoded sound map as described later. The complex-valued random mask acts as a special covering image with a virtual speckle pattern in terms of cryptography. In cyberspace we can numerically simulate such a scheme under an environment of MATLAB6.0. The complex-valued random mask can be generated by use of MATLAB build-in function *rand*(●) or *randn*(●) for the amplitude and phase parts, respectively. The amplitude of the mask is with Gaussian distribution and the phase is with uniform distribution.

Next, we import a digital audio signal with another MATLAB build-in function *waveread*(●) that can be applied to read a digital audio file with an extension name ".wav" into an one dimensional (1-D) array. Further, we reorder this 1-D array as a data block with a size of 65536 bytes. If the entries of the array were less than 64K, we would fill in zeros for the rest. Consequently, we can rearrange the 1-D array wherein digital audio signal (DAS) is stored as a two-dimensional (2-D) array with a size of 256×256. In other words,

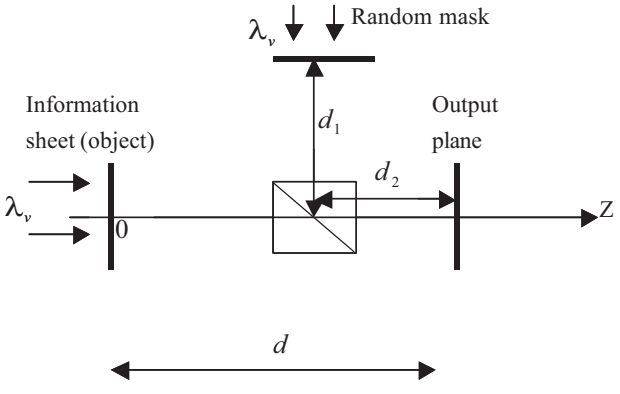


Fig. 1. Virtual-optics based scheme for enciphering digital audio signal.

we have encoded 1-D digital audio signal with a 2-D matrix that is referred to as 2-D sound map (2SM). The 2SM is thus stored in the memory as an information sheet and can be dealt with as an ordinary image.

As illustrated in fig. 1, both the 2SM (information sheet) and the RM are envisioned being "illuminated" from the left of the information sheet and from above the random mask with a virtually coherent light λ_v. The DFD transformation of the 2SM and the RM, propagating through a spatial distance d and $d_1 + d_2$, respectively, can be evaluated with following expressions according to the theory of Fourier optics [15]

$$U_{2SM}(m, n, d) = \frac{1}{\mathrm{i}\lambda_v d} \exp\left[\mathrm{i}\,\frac{\pi}{\lambda_v d}\,(m^2\,\Delta\xi^2 + n^2\,\Delta\eta^2)\right] \times \sum_{k=0}^{N-1}\sum_{l=0}^{N-1}\left\{U_{2SM}(k, l, 0)\exp\left[\mathrm{i}\,\frac{\pi}{\lambda_v d}\,(k^2\,\Delta x^2 + l^2\,\Delta y^2)\right]\right\} \times \exp\left[-\mathrm{i}2\pi\left(\frac{km}{N} + \frac{ln}{N}\right)\right] \quad (1)$$

and

$$U_{RM}(m, n, d_1 + d_2) = \frac{1}{\mathrm{i}\lambda_v (d_1 + d_2)} \exp\left[\mathrm{i}\,\frac{\pi}{\lambda_v (d_1 + d_2)}\,(m^2\,\Delta\xi^2 + n^2\,\Delta\eta^2)\right] \times \sum_{k=0}^{N-1}\sum_{l=0}^{N-1}\left\{U_{RM}(k, l, 0)\exp\left[\mathrm{i}\,\frac{\pi}{\lambda_v (d_1 + d_2)}\,(k^2\,\Delta x^2 + l^2\,\Delta y^2)\right]\right\} \times \exp\left[-\mathrm{i}2\pi\left(\frac{km}{N} + \frac{ln}{N}\right)\right], \quad (1)$$

where

$$x = k\,\Delta x, \qquad y = l\,\Delta y \quad k, l = 0, 1, \ldots N-1, \quad (2)$$

$$\xi = m\,\Delta\xi, \qquad \eta = n\,\Delta\eta \quad m, n = 0, 1, \ldots N-1, \quad (3)$$

$$f_x = m\,\Delta f_x = m\,\frac{\Delta\xi}{\lambda z}, \qquad f_y = n\,\mathrm{d}\eta = m\,\frac{\Delta\eta}{\lambda z},$$

$$z = \begin{cases} d, & \text{for 2SM} \\ d_1' + d_2', & \text{for RM} \end{cases}. \quad (4)$$

From above relations, we can see that DFD transform of 2SM and RM can be viewed as Fourier transform of the complex amplitude $U(k, l, z)|_{z=d,\, z=d_1+d_2}$ multiplied by a quadratic factor $\exp\left[\mathrm{i}\,\frac{\pi}{\lambda z}\,(k^2\,dx^2 + l^2\,\Delta y^2)\right]\Big|_{z=d,\, z=d_1+d_2}$. The Fourier transformation must take values at the spatial frequency plane, namely, $f_x = \xi/\lambda z$, $f_y = \eta/\lambda z$ to keep correct scale on the observation plane. In the light of well-known Shannon sampling theorem, we have

$$\Delta x = \frac{1}{2W}, \quad (5)$$

where $2W$ is the bandwidth of the spectrum of complex amplitude of the 2SM for the case of propagating from object plane to the interference plane, and therefore

$$\Delta x = \frac{1}{N\,\Delta f_x} = \frac{\lambda d}{N\,\Delta\xi}\,, \qquad \Delta y = \frac{1}{N\,\Delta f_y} = \frac{\lambda d}{N\,\Delta\eta}\,. \tag{6}$$

In order to facilitate the display, we keep identical the maximum size of information sheet and the its Fresnel diffraction pattern. In this case we can easily obtain, from eq. (2)–(6), that the maximum sizes should satisfy:

$$x_{\max} = y_{\max} = \xi_{\max} = \eta_{\max} = \sqrt{\lambda z / N}\,, \tag{7}$$

where N is total numbers of sample points in spatial domain and in frequency domain. In our case, N is taken as 256.

The synthetic complex amplitude at the interference plane is coherent superposition of the DFD pattern of the 2SM, propagating through a distance d, with the DFD pattern of the random mask propagating through a distance $d_1 + d_2$, which can be envisioned as a virtual speckle pattern. Therefore, the complex distribution at output plane (see fig. 1) can be expressed as

$$\begin{aligned} &U_{\text{out}}(m, n; d, d_1, d_2, \lambda_v) \\ &= U_{\text{RM}}(m, n; d_1 + d_2, \lambda_v) + U_{\text{2SM}}(m, n; d, \lambda_v)\,. \end{aligned} \tag{8}$$

The synthesized field above is stored in a complex-valued fashion and thus taken as embedded information sheet, namely, an enciphered signal. The enciphered 2SM (eq. (8)) can be sent via an ordinary communication link to authorized user. It can be easily seen that parameters d, d_2, d_2, λ_v, in conjunction with the random mask of this virtual scheme can all be individually utilized or incorporated with one and others for the design multiple-locks and multiple-keys during the encryption process. It would be extremely difficult for unauthorized third party to decipher the hidden host data without correct keys.

For an authorized user, he is able to decipher the encrypted 2SM (information sheet) and thus the host digital audio signal by following three steps: (1) taking the DFD transformation of the RM with a diffraction distance $d_1 + d_2$ and with correct wavelength λ_v, then subtracting the diffraction pattern from enciphered data; (2) taking the inverse DFD transformation with a diffraction distance d to correctly obtain deciphered 2-D sound map; (3) converting the 2-D sound back into recovered digital audio signal.

3. Numerical experiment and results

Numerical experiments are performed to demonstrate the validity of this scheme. As an example we take either λ_v or $d_1 + d_2$ in conjunction with a random mask as tools for the design of multiple-locks and multiple-keys in the course of enciphering digital audio signal. We also analyze the sensitivities of decryption by taking into account the deviations from correct values from those two parameters. Fig. 2 shows the wave of DAS, whose converted version of the

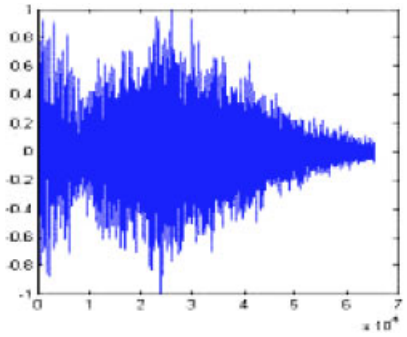


Fig. 2. Digital audio signal (DAS).

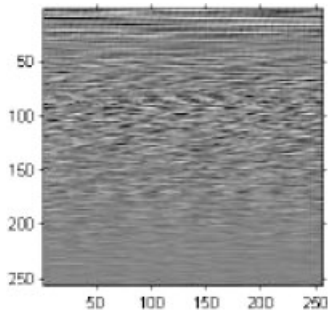


Fig. 3. 2-D sound map (2SM) of digital audio signal.

2SM is shown in fig. 3. In the simulation process we randomly select a virtual wavelength (e.g. $\lambda_v = 600$ nm). Other geometric parameters are chosen as: $d = 1.0$ m, and $d_1 + d_2 = d$ for convenience. The size of the information sheet (i.e. 2-D sound map) is controlled to be identical to that of Fresnel domain with $\sqrt{\lambda\,\mathrm{d}N} = 12.4$ mm.

The complex-valued random mask used as a covering image is shown in fig. 4, which looks like a "diffused screen". Enciphered data is illustrated in fig. 5a)–b), where a) represents enciphered audio signal, and b) shows embedded data in its 2SM version.

With correct keys of optical parameters (wavelength and Fresnel distance) and correct random mask code, one is able to obtain the deciphered DAS

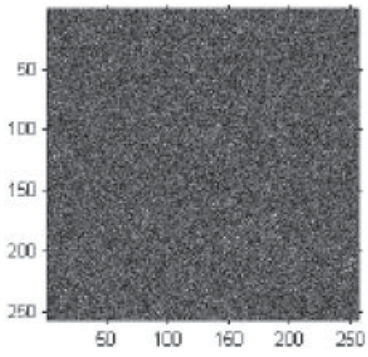


Fig. 4. Complex-valued random mask used for encryption.

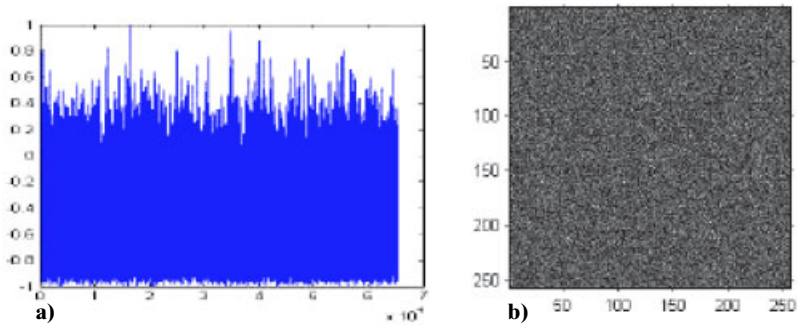


Fig. 5. Enciphered data of digital audio signal: a) encrypted digital audio signal (DAS); b) encrypted 2-D sound map (2SM) version of a).

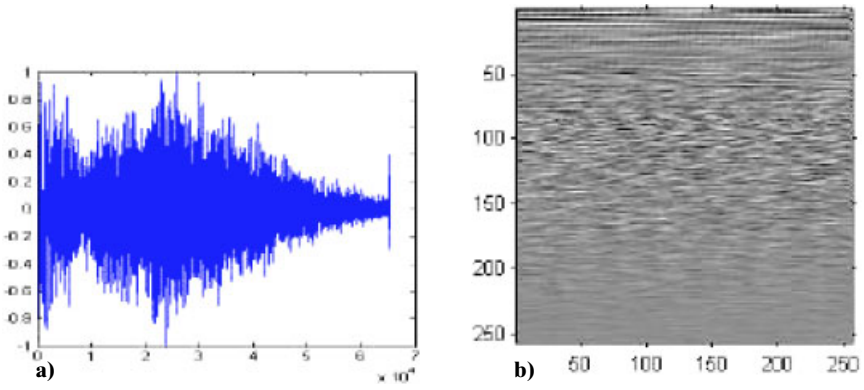


Fig. 6. Deciphering result with correct parameters and correct random mask. a) Deciphered DAS; b) Deciphered 2SM.

and 2SM as illustrated in fig. 6. With correct virtual wavelength λ_v (600 nm) and RM keys, but there are different amount of discrepancies in the diffraction distance key $d_1 + d_2 = d_{\text{key}}$, deciphering results are shown in fig. 7.

Fig. 7a) and b) show numerical experiment results (deciphered DAS and 2SM) where indicates that wavelength and RM keys are correctly used, but there is a discrepancy amounted to 0.05 mm in diffraction distance; c) (2SM) and d) (DAS) show the results where wavelength and RM keys are correct, but there is a deviation amounted to 0.01 mm in diffraction distance. We have

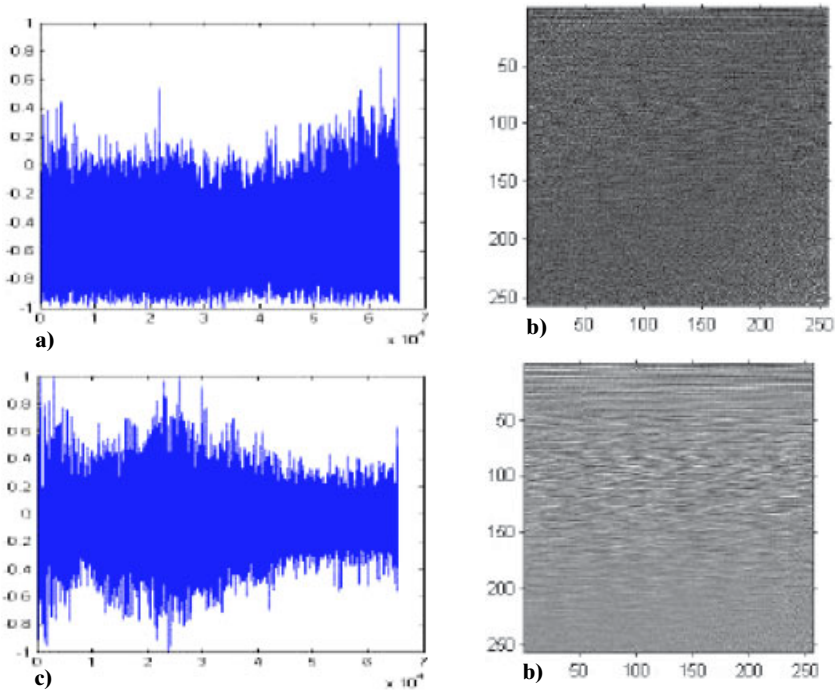


Fig. 7. Deciphering results (DAS and 2SM) with correct λ_v but with discrepancies in d_{key}: a) and b) with discrepancy of 0.05 mm; c) and d) with discrepancy of 0.01 mm.

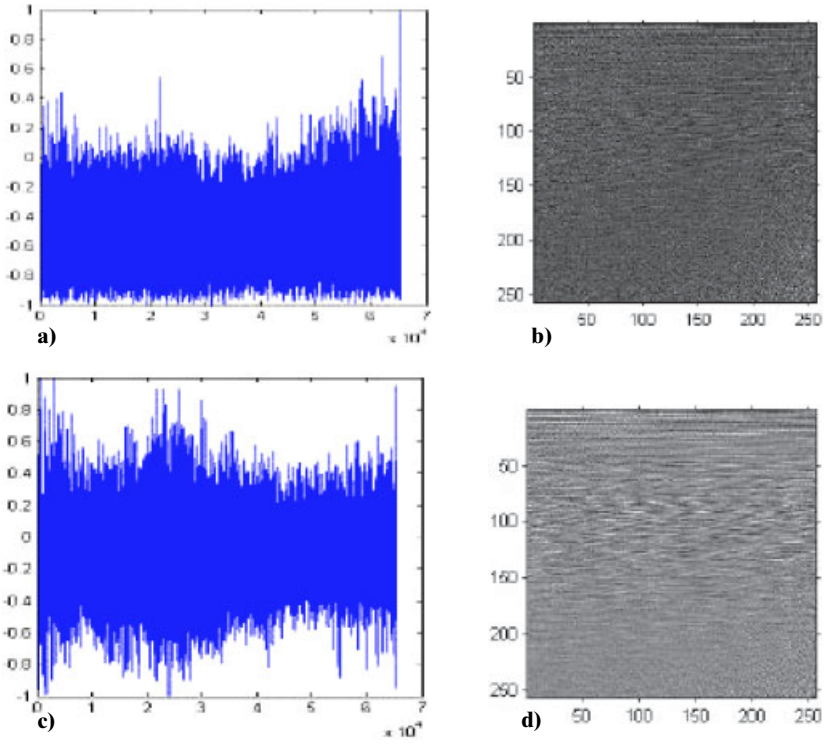


Fig. 8. Deciphered result with correct d_{key}, but with discrepancies in λ_v: a) and b) with discrepancy of 0.03 nm; c) and d) with discrepancy of 0.01 nm.

found out, from the experiment, that when the discrepancy is less than 0.01 mm, the recovered audio signal is quite close to that of host signal (see discussion in the next section). So the sensitivity of diffraction distance for deciphering an audio signal is about 0.01 mm for the given wavelength. The error in distance is 0.01 mm out of a total distance of 1 meter. This is a fractional accuracy of 100 000.

Now we keep virtual diffraction distance and RM keys with correct values and deliberately introduce discrepancies in virtual wavelength to test the behavior of the process of audio signal decryption. Numerical experiment results are shown in fig. 8.

Fig. 8a) and b) depict deciphering results (DAS and 2SM) with correct diffraction distance (1.0 m) and RM keys, but with 0.03 nm drift in virtual wavelength; c) and d) give the results with a drift 0.01 nm, form which one can see the host audio wave has appeared. So, we conclude that the sensitivity of virtual wavelength for deciphering an audio signal should be at 0.01 nm for given diffraction distance. Again, if we take a fractional measure the error in virtual wavelength is 0.01 nm out of a total wavelength of 600 nm. This is a fractional accuracy of 60 000. Therefore the total key dimension in this scheme is a large number, aside from that provided by the RM. Actually another huge dimension of the keys comes from random mask. For example, if the RM is set to be 8-bit real valued random noise with 256×256 pixels, thus the total number of possible masks could be $(2^8)^{256\times256}$, which is a huge number. It would be really difficult for any digital methods to successfully attack the encrypted information. And comparing the possibility to use real optical system to attack the digitally encoded information, it is even more difficult than digital methods, because even if all the optical parameters and the physical parameters are known, the tolerance of 0.01 nm might be too small to realize or the virtual wavelength may not exit in real world. We have observed that both parameters are sensitive for the audio signal encryption and decryption. This virtual scheme can introduce huge dimension for the multiple-lock and multiple-key design. In addition, a complex-valued random mask also plays a significant role for increasing the security level in the process of enciphering digital audio signal.

4. Analysis of decryption capability

The correlation between original signal and decrypted signal is a measure to estimate the degree of their similarity. In order to quantitatively evaluate the similarities between decrypted audio signals and original signal, we carry out correlation analysis by considering the decrypted signals with different conditions in order to verify the capability of this scheme for data decryption.

The cross-correlation of stochastic processes $x(t)$ and $y(t)$, real or complex, is defined as [16]

$$R_{xy}(t_1, t_2) = E\{x(t_1)\, y^*(t_2)\}\,, \tag{9}$$

where the conjugate term is associated with the second variable in $R_{xy}(t_1, t_2)$ and $E\{\cdot\}$ is an expectation operator. The cross-covariance $C_{xy}(t_1, t_2)$ of processes $x(t)$ and $y(t)$ is the covariance of the random variables $x(t_1)$

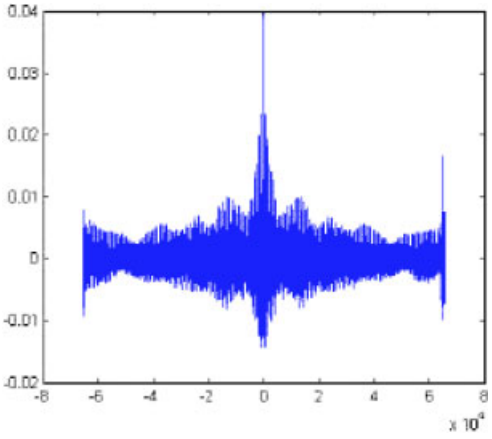


Fig. 9. Output correlation between original audio signal and decrypted signal with correct keys (wavelength, diffraction distance, and random mask).

and $y(t_2)$

$$C_{xy}(t_1, t_2) = R_{xy}(t_1, t_2) - |\eta_x(t)|\,|\eta_y(t)|\,. \tag{10}$$

In the above $\eta_x(t) = E\{x(t)\}$ is the mean of $x(t)$ and $\eta_y(t) = E\{y(t)\}$ is the mean of $y(t_2)$.

The ratio

$$r(t_1, t_2) = \frac{C_{xy}(t_1, t_2)}{\sqrt{C_{xx}(t_1, t_1)\; C_{yy}(t_2, t_2)}} \tag{11}$$

is the correlation coefficient of the processes $x(t)$ and $y(t_2)$.

Fig. 9 shows the correlation between original signal and decrypted signal with correct keys (wavelength, diffractive distance, and random mask), where absolute value of the correlation coefficient is $|r| = 0.0502$. In this case we are able to almost recover the signal as illustrated as fig. 6a).

Fig. 10a) illustrates the output correlation for the audio signal decryption when existing 0.01 nm wavelength drifting. In this case absolute value of the correlation coefficient is $|r| = 0.0435$ and recovered signal is shown as fig. 8c). The output correlation for the decryption with a wavelength discrepancy of 0.03 nm is shown in fig. 10b) where absolute value of the correlation coefficient is $|r| = 0.0238$. Decrypted signal in this case has been shown in fig. 8a). From experiment we have observed that when absolute value of the correlation coefficient decreases to 0.0238 the audio signal decryption is failed whereas the correlation coefficient approaches 0.0435 decrypted audio signal can be detected successfully. In other words, the recovered signal is immersed into the noise when $|r| \leq 0.0238$.

Fig. 11a) shows output correlation for the audio signal decryption when existing a discrepancy 0.01 mm in diffractive distance. The absolute value of correlation coefficient for this case is $|r| = 0.0478$, leading to a decrypted signal as shown in fig. 7c). Output correlation

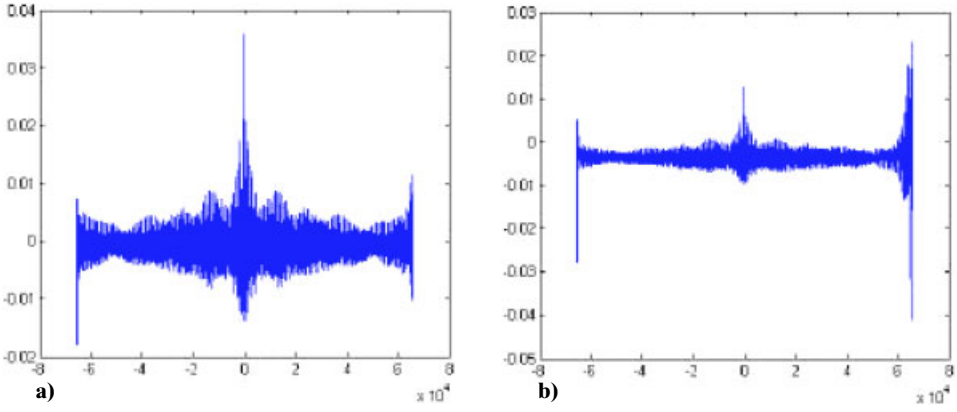


Fig. 10. Output correlation between decrypted signals and original signal: a) decrypted signal with wavelength drifting 0.01 nm; b) decrypted signal with wavelength drifting 0.03 nm.

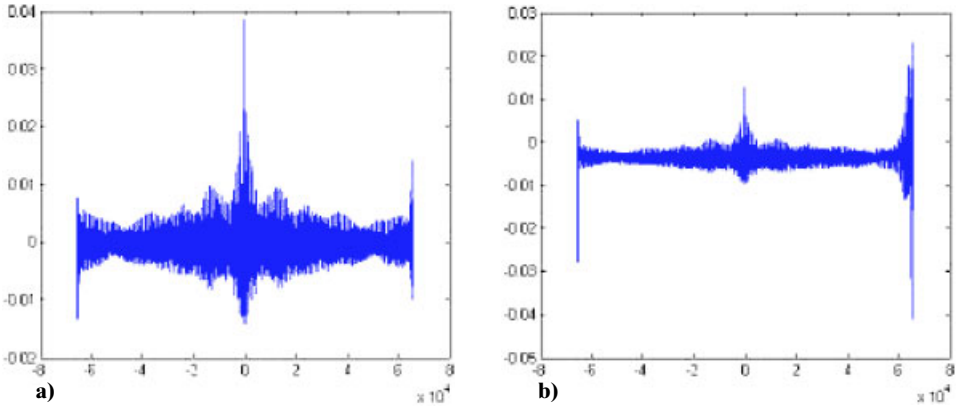


Fig. 11. Output correlation between decrypted signals and original signal: a) decrypted signal with distance drifting 0.01 mm; b) decrypted signal with distance drifting 0.05 mm.

for the audio signal decryption with a discrepancy 0.05 mm in diffractive distance is illustrated fig. 11b) where absolute value of the correlation coefficient for this case is $|r| = 0.0238$. Decrypted signal corresponds to this case has been shown in fig. 7a).

5. Conclusion

In conclusion, we have presented a new virtual scheme for enciphering and deciphering digital audio signal. With this method, the host audio signal is converted into a 2-D sound map in which original digital audio signal has been encoded. The 2-D sound map can be dealt with as an image. The parameters of virtual optical scheme such as virtual wavelength and virtual diffraction distance in conjunction with a random mask have been explored as tools for the design of multiple-locks and multiple-keys in the course of data encryption and decryption. Numerical experiment and results have been presented to show the effectiveness of proposed methodology. In addition, a correlation analysis of decrypted audio signal comparing with original audio signal is also performed to show the parameters sensitivities of proposed virtual scheme and its capability for the audio signal decryption. It has been shown that huge dimension of keys could be introduced by virtual wavelength and spatial distance incorporated with a complex-valued random mask. The fractional accuracies of virtual wavelength used for deciphering audio signal is at amount 60 000 for given wavelength 600 nm, and fractional accuracies of spatial distance is at 100 000 for a given diffraction distance 1 meter. More practical implementation of proposed scheme is under study and relevant results will be reported soon.

Acknowledgements. The sponsorship from the NLPR, Chinese Academy of Science through an "Opening Research Project" is greatly acknowledged. The authors are also grateful to the financial support by Research Grants Council (RGC) of Hong Kong through a research project HKUST6015/02E.

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